Critical Care Nephrology

65

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CHAPTER OUTLINE

KEY POINTS

- In the setting of acute liver failure, continuous renal replacement therapt (CCRT) is the preferred renal replacement therapy (RRT) modality to mitigate the risks of cerebral edema and herniation.
- Permissive hypercapnia can occur with lung-protective low tidal volume ventilation, indicated for the management of the acute respiratory distress syndrome; with concomitant acute kidney injury (AKI), a severe mixed acidosis may arise.
- There is no benefit to early goal-directed therapy for the management of sepsis, either with regard to mortality or the prevention of AKI.
- The use of balanced crystalloids for resuscitation may reduce the risk of AKI.
- Although there is no definitive evidence favoring a particular RRT modality, it is reasonable to use intermittent and continuous modalities in the context of the patient's hemodynamic status.
- The optimal timing of dialysis initiation in severe AKI remains controversial; although the preemptive initiation of RRT in the absence of any emergent indications has conceivable benefits, it may expose patients who might otherwise recover kidney function without a need for RRT to undue risk.
- In critically ill patients with AKI receiving CRRT, effluent flow should be targeted to 25 mL/kg/hour.
- In critically ill patients with AKI, frequent and careful review of medications is warranted, with close attention to the avoidance of nephrotoxins and agents that accumulate in AKI while tailoring dosing and frequency of medication administration to the RRT modality in use.

Studies now suggest that up to 60% of patients in intensive care units (ICUs) experience an episode of acute kidney injury (AKI).[1-3] In the most severe cases, renal replacement therapy (RRT) is commenced. Short-term mortality among critically ill patients with acute AKI who receive RRT is in excess of 50%[4] and, among survivors, longevity and health-related quality of life are often poor.[5] AKI, as well as electrolyte and acid-base abnormalities, occurs on the background of multiple conditions that necessitate critical care. In this chapter, we first discuss AKI in the context of conditions that are frequently encountered in the ICU—acute liver failure, acute respiratory distress syndrome, acute decompensated heart failure, and sepsis. This is followed by a discussion of fluid management and its relevance to the care of patients with AKI. We then review the principles of RRT delivery to critically ill patients with AKI. We do not discuss acute brain injury in the ICU setting, which is associated with dysnatremias (including the syndrome of inappropriate secretion of antidiuretic hormone, cerebral salt wasting, and central diabetes insipidus). The reader is referred to Chapter 15 for further discussion of these disorders.

ACUTE KIDNEY INJURY IN THE CONTEXT OF CRITICAL ILLNESS

LIVER FAILURE

In the setting of acute liver failure, a major cause of death is cerebral edema, resulting in elevated intracranial pressure and, if severe or sustained, brain stem herniation.[6] Elevated intracranial pressure occurs in up to 35% of patients with grade III and 75% of patients with grade IV encephalopathy.[7] Continuous RRT (CRRT) is often used for meticulous control of volume status in these patients, given the high risk of cerebral edema in hepatic failure and frequent high obligate intake in the form of infusions (*N*-acetylcysteine, vasopressors, and blood products, including fresh-frozen plasma). CRRT can also be used to manage hyponatremia, a common complication of liver failure that may further exacerbate cerebral edema.

Patients with acute liver failure are at risk of developing AKI from hepatorenal syndrome or acute tubular necrosis. Acute tubular necrosis may be caused by profound hypotension, concomitant sepsis (a common complication of acute liver failure), or ingestion of a substance that is both hepatotoxic and nephrotoxic (e.g., acetaminophen, *Amanita* mushrooms).

ACUTE RESPIRATORY DISTRESS SYNDROME

The mainstay of supportive care for patients with acute respiratory distress syndrome (ARDS) is low tidal volume, lung-protective ventilation, which has led to a significant reduction in mortality and is associated with decreased levels of proinflammatory cytokines.[8] As part of the lung-protective ventilation strategy, permissive hypercapnia is encouraged to minimize ventilator-associated lung injury. In the setting of normal kidney function, compensatory metabolic alkalosis will ensue. However, in the setting of impaired kidney function (acute or chronic), the metabolic acidosis that arises and the inability to compensate for associated respiratory acidosis may prompt the administration of bicarbonate-containing infusions or trigger the initiation of RRT for correction of the acid-base imbalance. In patients with severe ARDS in whom ventilation is markedly impaired because of alveolar injury (as reflected by a high dead space fraction and impaired respiratory carbon dioxide excretion), bolus doses of bicarbonate may worsen arterial pH through an abrupt increase in $Pa{CO_2}$ and conversion to carbonic acid.[9] Therefore, CRRT may be the preferred modality to correct pH slowly and compensate for respiratory acidosis. A fluid conservative management strategy has been associated with improved outcomes and no adverse kidney consequences, although patients with dialysis-requiring AKI at the time of study enrollment were excluded from the study.[10]

Studies in animal models have suggested that there is significant cross-talk between the kidney and lung—that is, injury to one organ results in injury to the other.[11-14] For example, kidney injury is associated with elevations of proinflammatory cytokines and worse lung injury in mice[15]; in one study, concentrations of interleukin-6 and interleukin-8 were higher in children with AKI after cardiopulmonary bypass surgery than in controls (children without AKI) and were associated with a longer duration of mechanical ventilation.[16]

HEART FAILURE

Because cardiac function affects kidney function and vice versa, the following five clinical subtypes of the cardiorenal syndrome (CRS) have been proposed:[17,18]

1. Acute CRS (type 1), in which acute worsening of heart function leads to AKI
2. Chronic CRS (type 2), in which chronic abnormalities in heart function result in impaired kidney function
3. Acute renocardiac syndrome (type 3), in which AKI precedes impaired cardiac function
4. Chronic renocardiac syndrome (type 4), in which CKD leads to impaired cardiac function
5. Secondary CRS (type 5), in which systemic conditions such as sepsis result in simultaneous impairment in cardiac and kidney function

With type 1 CRS in particular, volume overload and consequent venous congestion may result in kidney dysfunction via several mechanisms, including profound hypotension compromising kidney perfusion, kidney venous hypertension, the development of ascites with intraabdominal hypertension, and perhaps inflammation.[19] Consequently, the use of diuretics to improve volume overload in acute decompensated heart failure may actually improve kidney function by relieving renal venous hypertension. On the other hand, excessive diuresis may lead to volume depletion and cause or exacerbate AKI.

There has been significant interest in the optimal management of volume overload in the context of acute decompensated heart failure. Several studies have compared the use of continuous versus intermittent bolus dosing of loop diuretics[20,21] on the basis of the hypothesis that continuous infusion results in more effective diuresis by avoiding periods of "rebound" sodium retention between bolus doses. However, the incremental benefit of continuous infusions over bolus dosing has not been demonstrated to date.

Extracorporeal ultrafiltration has been proposed as an alternative to diuretic management for volume overload in acute decompensated heart failure and has been tested in a number of randomized clinical trials. The Cardiorenal Rescue Study in Acute Decompensated Heart Failure (CARESS-HF) trial compared the safety and efficacy of ultrafiltration (with a target fluid removal of 200 mL/hour) to stepped pharmacologic therapy (target urine output of 3–5 L/day) in patients with acute CRS type 1 (Fig. 65.1).[22] The primary endpoint was the bivariate change from baseline in serum creatinine concentration and body weight. The trial was terminated early owing to a lack of benefit in the ultrafiltration group (serum creatinine level increased slightly compared with the pharmacologic group, and weight change was similar in both groups), combined with an increased

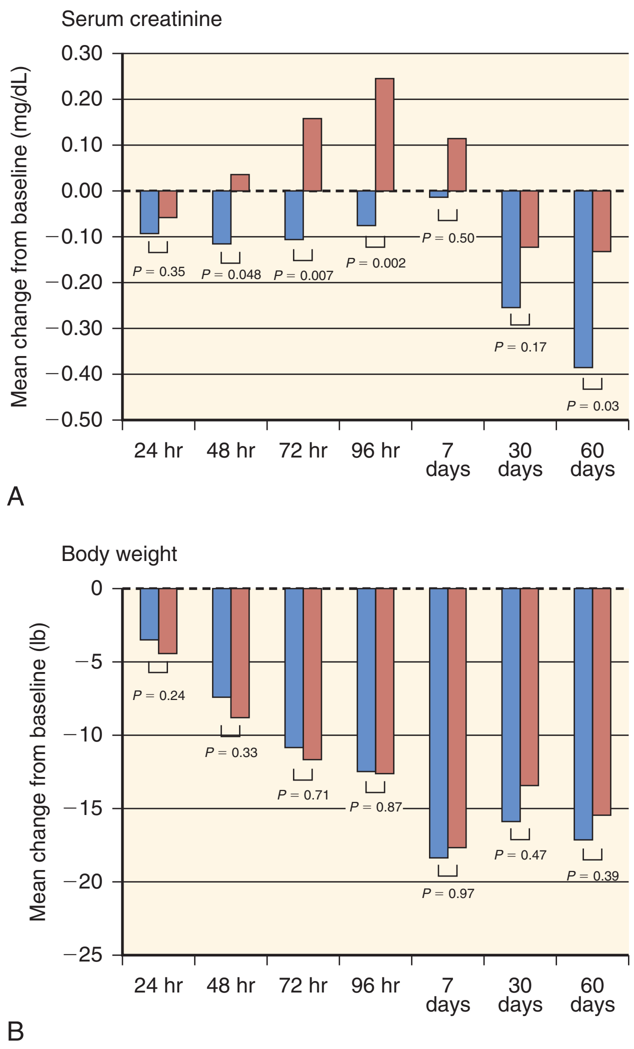

Fig. 65.1 Role of ultrafiltration in decompensated heart failure. Changes from baseline in (A) serum creatinine level and (B) body weight at various time points, according to treatment group. The *P* values were calculated with the use of a Wilcoxon test. (From Bart BA, Goldsmith SR, Lee KL, et al. Heart Failure Clinical Research Network: ultrafiltration in decompensated heart failure with cardiorenal syndrome. *N Engl J Med.* 2012;367:2296–2304.)

risk of adverse events, including bleeding and catheter-related complications. On the basis of this experience, ultrafiltration is no longer favored as a first-line therapy for patients with acute decompensated heart failure. Nonetheless, ultrafiltration may be required in patients with acute decompensated heart failure that fails to respond to maximal doses of intravenous diuretics and other pharmacotherapy (e.g., cardiac inotropic agents), particularly in the presence of acute kidney injury (CKD) and/or acute kidney injury (AKI).

SEPSIS

AKI is a common complication of sepsis and occurs via multiple mechanisms, including hypotension leading to hypoperfusion, inflammation, and oxidative stress.[23,24] There are no specific treatments for AKI or for sepsis itself, apart from source control and treatment with appropriate antimicrobials. With regard to inflammatory mediators in sepsis, there has been interest in the use of high-volume hemofiltration and adsorptive (e.g., polymyxin B) columns to remove proinflammatory cytokines, but studies to date have not shown any benefit.[25,26]

With regard to the early management of sepsis, publication of the pivotal study by Rivers and colleagues[27] has stimulated a major shift in clinical practice toward early "goal-directed therapy," in which inotropes and packed red blood cell transfusions are used in addition to early volume resuscitation and vasopressors in patients with septic shock. However, the results of three subsequent large randomized clinical trials have suggested no benefit of early goal-directed therapy over standard therapy.[28-30] The ProCESS (Protocol-Based Care for Early Septic Shock) trial randomly allocated 1341 patients in the United States to protocol-based early goal-directed therapy, protocol-based standard therapy, or usual care. There was no difference in 60-day, 90-day, or 1-year mortality among the treatment arms (60-day mortality 21% in the protocol-based early goal-directed therapy arm, 18.2% in the protocol-based standard therapy arm, and 18.9% in the usual care arm). There was also no impact on the rate of AKI after enrollment (40.3% in the protocol-based early goal-directed therapy arm, 34.9% in the protocol-based standard therapy arm, and 38.1% in the usual care arm).[31] Patients in the protocol-based standard therapy and usual care arm received fewer packed red blood cell transfusions and less dobutamine than patients in the other two arms, suggesting that these interventions are of limited benefit in a general population with sepsis. These findings have been further supported by the ProMISe (Protocolised Management in Sepsis) and ARISE (Australasian Resuscitation in Sepsis Evaluation) trials. An ongoing clinical trial, CLOVERS (Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis), is comparing the impact of early fluid resuscitation versus early vasopressor use in patients with septic shock.[32]

FLUID MANAGEMENT AND ITS IMPACT ON ACUTE KIDNEY INJURY

There has been an increasing appreciation of the potentially deleterious effects of fluid overload in critically ill patients, particularly in the context of AKI. Fluid excess has been associated with a number of adverse consequences, including decreased gastrointestinal absorption and impaired wound healing.[33] In patients with AKI, fluid overload has been independently associated with an increased risk of new sepsis, as well as with increased short- and long-term risks of death.[34-37] However, the observational nature of these studies makes it difficult to determine if the relationship between fluid overload and mortality is causal. Specifically, it is unclear whether fluid overload directly mediates adverse outcomes or if sicker patients who are hemodynamically unstable—and who have an inherently worse prognosis—are more likely to receive fluid boluses and other infusions that generate a more positive fluid balance. Furthermore, total body fluid overload may dilute serum creatinine concentration and mask AKI.[38,39]

An adverse consequence of fluid overload that has gained significant interest is intraabdominal hypertension and the abdominal compartment syndrome.[40] Intraabdominal hypertension is defined as an intraabdominal pressure greater than 12 mm Hg. This pressure is typically measured by instilling a fixed volume of water (30 mL) into the urinary bladder via a Foley catheter and using pressure tubing to transduce a bladder pressure. The abdominal compartment syndrome is defined as an intraabdominal pressure greater than 20 mm Hg with associated end-organ dysfunction. Intraabdominal hypertension may mediate AKI through direct compression of the inferior vena cava, which results in impaired venous return and venous stasis throughout the abdominal cavity, including the renal veins.[41,42] Impaired venous return leads to decreased cardiac output and increased sympathetic and renin-angiotensin-aldosterone system (RAAS) signaling, resulting in renal vasoconstriction. The result is a functional prerenal state, characterized by low urinary sodium concentration and oliguria. Decompression of the abdominal compartment (typically via a surgical approach) may lead to an improvement of kidney function.[40] However, identification of patients whose kidney function will benefit from decompression remains elusive.

There has also been significant interest in the type of fluid administered and its impact on kidney function. It has been suggested that chloride-rich solutions may cause renal vasoconstriction and exacerbate renal medullary hypoxia.[43] In a small crossover study of human volunteers, administration of chloride-rich solutions was associated with greater fluid retention and reduced kidney perfusion than administration of balanced salt solutions.[44] A number of observational studies have suggested that the use of chloride-rich solutions is associated with a higher risk of AKI than balanced salt solutions.[45,46] The largest of these studies was a single-center, prospective, open-label sequential study.[46] During the 6-month control period, patients received normal saline (chloride concentration, 154 mmol/L, or chloride-rich colloids) for resuscitation. During the intervention period that commenced six months thereafter, patients received Plasma-Lyte (chloride concentration, 98 mmol/L) or chloride-restricted colloids for resuscitation. The incidence of AKI was 14% in the control period and 8.4% in the intervention period ($P < .001$). This was accompanied by a lower use of RRT (10% and 6.3%; P = .005) during the control and intervention periods, respectively). Challenges to the interpretation of these findings include the larger than expected effect size, concurrent changes made to other aspects of fluid management, and other practice changes that may have reduced the risk of AKI. One potential explanation for this difference is that the concomitant hyperchloremic metabolic acidosis from chloride-rich fluid administration may have triggered earlier RRT initiation.

The Saline vs Plasma-Lyte 148 for ICU fluid Therapy (SPLIT) trial[47] compared 0.9% Saline vs Plasma-Lyte 148 using a double-blind, double-crossover cluster design in four ICUs in New Zealand. Participating units were randomized to administer one of the study solutions to all admitted patients in two alternating 7-week blocks. Plasma-Lyte 148 did not lower the risk of acute kidney injury, the primary outcome of the trial, nor did it affect the receipt of RRT or any other marker of ICU resource utilization. A key limitation of this trial was the high number of patients who were enrolled postcardiac surgery, which may limit generalizability. More recently, two very large, pragmatic, single-center clinical trials have been published: SALT-ED (Saline against Lactated Ringer's or Plasma-Lyte in the Emergency Department, N= 13,347)[48] and SMART (Isotonic Solutions and Major Adverse Renal Events Trial, N = 15,082).[49] Patients admitted to the emergency department or ICU, respectively, at Vanderbilt University Medical Center were allocated to receive a balanced salt solution (selected by the clinician—e.g., lactated Ringer's or Plasma-Lyte) or normal saline in alternating months. Although clinicians could choose to break the allocation and prescribe the opposite fluid, protocol compliance was excellent, at greater than 88% across the two arms of both studies. Both SALT-ED and SMART demonstrated a reduction in a composite endpoint of Major Adverse Kidney Events at 30 days (MAKE-30, which is comprised of a persistent doubling of serum creatinine level, new RRT, or death) with the use of balanced salt solutions (14.3% vs. 15.4% in SMART, P = 0.04; 4.7% vs. 5.6% in SALT-ED, P = .01). Thus these two studies add strength to the argument that there is no benefit to the use of normal saline and possibly some potential harm. The ongoing Plasma-Lyte 148 versUs Saline (PLUS) trial, with a planned recruitment of 8800 critically ill patients and a primary outcome of 90-day mortality, may provide more definitive guidance on this question.[50]

RENAL REPLACEMENT THERAPY IN THE INTENSIVE CARE UNIT

GOALS OF CARE

Severe AKI frequently prompts discussions about the initiation of RRT. The broad goal of RRT revolves around management of the fluid and metabolic abnormalities of AKI in a manner that facilitates and promotes recovery from the underlying illness. RRT should ideally be performed with minimal disruption to the broader care received by the patient. One may debate whether RRT is merely supportive or confers an additional therapeutic benefit that may enhance survival. At present, there are no data on any specific RRT-related maneuver that improves patient survival. A further goal of RRT is maximizing the likelihood of recovery of kidney function among surviving patients. This objective is driven by the growing recognition that AKI survivors are at high risk for CKD.[51,52] RRT protocols should be designed to minimize iatrogenic injury to the kidney (e.g., injury due to hypotension and/or hypovolemia) in the hope that dialysis independence can be achieved, with an eventual return to the level of kidney function that preceded the acute illness (or almost to this level).

INDICATIONS FOR COMMENCING RENAL REPLACEMENT THERAPY

The decision to initiate RRT is unambiguous when AKI is complicated by hyperkalemia, pulmonary congestion refractory to medical maneuvers, or concomitant intoxication with a dialyzable toxin.[53] However, even in the absence of a life-threatening AKI complication, some have argued in favor of a strategy of early or preemptive RRT initiation. Proponents of this approach have cited the mitigation or reversal of volume overload and tempering the toxicity of uncontrolled uremia as justifications for the earlier initiation of RRT.[54,55] On the other hand, a preemptive approach to RRT initiation has conceivable shortcomings. Even in patients with severe AKI, spontaneous kidney recovery is frequently observed, and the widespread adoption of earlier RRT could result in the potentially unnecessary exposure of patients to the risks associated with vascular access placement and the RRT procedure itself (e.g., hypotension, arrhythmia, electrolyte abnormalities, ineffective antibiotic concentrations). Furthermore, if a strategy of earlier RRT initiation does not confer any meaningful patient benefits, adoption of a more conservative approach to RRT initiation and commencing RRT only when an overt complication develops could lead to resource savings.

The controversy surrounding the optimal circumstances for RRT initiation in AKI was addressed in two recently completed randomized trials.[56,57] The Artificial Kidney Initiation in Kidney Injury (AKIKI) trial was a multicenter trial of 620 critically ill adults with Kidney Disease Improving Global Outcomes (KDIGO) stage 3 AKI conducted in France. The investigators hypothesized that a delayed strategy of RRT initiation (RRT only after 72 hours of persistent oligoanuria, serum urea > 40 mmol/L, or development of an AKI-related emergency) would confer better patient survival at 60 days compared with a strategy of early initiation, defined as commencing RRT within 6 hours of patients meeting criteria for stage 3 AKI. Patients in the delayed arm commenced RRT 55 hours later than those in the early arm but, importantly, only half the participants allocated to the delayed arm started RRT. The trial did not show an appreciable difference in mortality between the two strategies (60-day mortality in early and delayed groups was 48.5% and 49.7%, respectively; $P = .79$).

A single-center trial in Germany, Effect of Early vs Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury (ELAIN), was designed to test the hypothesis that an early strategy of RRT initiation confers improved survival. The trial recruited 231 patients with stage 2 AKI who had a blood neutrophil gelatinase-associated lipocalin (NGAL) greater than 150 ng/mL and at least one other element of severe illness (e.g., one or more of sepsis, catecholamine dependence, hypoxemia, fluid overload, or progressive nonrenal organ injury). In the early RRT arm, RRT was started within 8 hours of fulfilling KDIGO stage 2 AKI, whereas in the delayed arm, RRT was deferred until KDIGO stage 3 criteria were met, or if an absolute indication became apparent. All patients in the early initiation arm received RRT at a median of 6 (interquartile range [IQR], 4–7) hours from meeting stage 2 AKI criteria; almost all (91%) patients in the late initiation arm attained a trigger for RRT initiation, which in most cases was achievement of KDIGO stage 3 AKI. These individuals started RRT at a median of 25.5 (IQR, 18.8–40.3) hours from reaching stage 2 AKI. Mortality was significantly lower in the early initiation arm compared with the delayed initiation arm (39.3% vs. 54.7%; $P = .03$; hazard ratio [HR], 0.66; 95% confidence interval [CI], 0.45–0.97). Secondary endpoints such as duration of RRT, mechanical ventilation days, and length of hospitalization were all markedly lower in the early arm. Table 65.1 provides a more detailed summary of the AKIKI and ELAIN trials.

The divergent findings in the AKIKI and ELAIN trials have not settled the controversy surrounding the optimal timing of RRT initiation in AKI. Moreover, both trials were

Table 65.1 Comparison of Artificial Kidney Initiation in Kidney Injury and Effect of Early Versus Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients With Acute Kidney Injury

Parameter	AKIKI	ELAIN
Principal hypothesis	Delayed RRT reduces 60-day mortality by 15%	Early RRT reduces 90-day mortality by 18%
No. of patients enrolled	620	231
Centers	31	1
Age	66	66
SOFA	11	16
CKD, %	10	43
Mechanical ventilation, %	86	88
Pressor requirement, %	85	88
Septic shock, %	67	32
Criteria for early RRT	KDIGO stage 3 AKI	KDIGO stage 2 AKI
Criteria for delayed RRT	Clinical indications	KDIGO stage 3 AKI
sCr at RRT initiation in early group (SD), mg/dL	1.9 (0.6)	3.3 (1.4)
sCr at RRT initiation in delayed group (SD), mg/dL	2.4 (1.0)	5.3 (2.3)
RRT modality	IHD, SLED, or CRRT	CVVHDF only for 7 days
Received RRT in early arm, %	98	100
Received RRT in delayed arm, %	51	91

AKI, acute kidney injury; *AKIKI,* Artificial Kidney Initiation in Kidney Injury; *CKD,* chronic kidney disease; *CRRT,* continuous renal replacement therapy; *CVVHDF,* continuous venovenous hemodiafiltration; *ELAIN,* Effect of Early vs Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury; *IHD,* intermittent hemodialysis; *KDIGO,* Kidney Disease Improving Global Outcomes; *RRT,* renal replacement therapy; *sCr,* serum creatinine; *SD,* standard deviation; *SLED,* sustained low-efficiency dialysis; *SOFA,* Sequential Organ Failure Assessment Score.

likely underpowered for the detection of a realistic difference in mortality that could be plausibly conferred by any RRT initiation strategy. Ongoing and recently completed studies will hopefully shed further light on this issue.[58,59]

RENAL REPLACEMENT THERAPY MODALITY

A variety of RRT modalities are available for the treatment of critically ill patients, each with a unique set of advantages and shortcomings (Table 65.2). However, no single modality has been shown to confer improved survival. As a result, logistic factors such as costs, local availability, and staff expertise can be justified as factors that determine modality choice.

INTERMITTENT HEMODIALYSIS

Intermittent hemodialysis (IHD) can be broadly defined as the application of a prescription designed for patients with end-stage kidney disease to individuals with AKI. However, some key modifications are needed for patients with AKI. Due to the lower blood flow rates typically achieved with temporary dialysis catheters and the high catabolic rates of critically ill patients, session durations up to 5 hours may be required.[60] Heparin is often omitted owing to the bleeding risks that are frequently seen in critically ill patients (see later discussion of anticoagulation). Blood flow ranges from 200 to 400 mL/min, with lower blood flows used in initial sessions for patients who are believed to be at risk for dialysis disequilibrium. In hemodynamically unstable patients, the use of a hypertonic dialysate sodium concentration (e.g., 145 mmol/L) may promote stability by enhancing fluid movement from the intracellular to extracellular space. Although hyperkalemia is often a trigger for RRT initiation, excessive potassium removal may result in hypokalemia-mediated arrhythmias. A dialysate potassium concentration less than 2 mmol/L should be avoided when possible.

To achieve euvolemia in a population that is generally fluid-overloaded,[35,37] the prescribed ultrafiltration volume must address the patient's expected intake from infusions and nutrition while achieving net fluid removal. The key challenge of IHD is the need to ultrafilter relatively large volumes in a short time in patients who are frequently hemodynamically compromised and have a tendency to poor vascular refilling from the interstitium. In addition to modification of the dialysate sodium concentration, intradialytic hypotension can be prevented or attenuated by lowering the dialysate temperature or initiating or escalating vasopressor doses. However, if a patient develops intradialytic hypotension or requires escalating doses of vasopressors to tolerate fluid removal via IHD, conversion to modalities (see later) that permit a slower rate of fluid removal should be considered. The introduction of more frequent IHD sessions (e.g., daily) or supplemental sessions devoted exclusively to ultrafiltration, may permit a lower ultrafiltration volume per session, thereby enabling the achievement of euvolemia with greater hemodynamic tolerability.

CONTINUOUS RENAL REPLACEMENT THERAPY

CRRT enables the slow removal of fluid and solutes using hemodialysis and/or hemofiltration, either exclusively or in combination (see later section on clearance mode). On a unit-time basis, CRRT is an inefficient form of RRT and thus would not be considered optimal when the goal is the rapid removal of a dangerous solute (e.g., potassium, ingested toxin). The efficacy of CRRT, and its putative benefits, is thus realized only when it is applied throughout the 24-hour period, with minimal interruption. Technical factors (e.g., frequent clotting) and time away from the ICU for procedures may hamper delivery that is truly continuous. Clinical practice guidelines from KDIGO have suggested using CRRT and intermittent forms of RRT in a complementary fashion.[61] A frequent approach is to use CRRT for patients who are hemodynamically unstable and IHD for patients who are more stable, with the understanding that intermodality changes will occur as the clinical picture evolves.

The administration of CRRT should be carefully protocolized. Ordering clinicians must determine the optimal intensity of RRT to be administered (see later section on dose and intensity). Fluid balance is frequently managed on an hourly basis; because volume overload is virtually ubiquitous in the critically ill patient with AKI, achievement of a net fluid removal mandates that hourly ultrafiltration exceeds the patient's overall fluid balance during the preceding hour. As an example, one may consider a desired net ultrafiltration goal of 50 mL/hour being prescribed to an anuric patient. If the patient received 30 mL from parenteral nutrition and 40 mL from a variety of infusions and lost 30 mL from a variety of postoperative drains (net balance + 40 mL), the

Table 65.2 Advantages and Shortcomings of Renal Replacement Therapy Modalities

Advantage or Shortcomings	Intermittent Hemodialysis	Sustained Low-Efficiency Dialysis	Continuous Renal Replacement Therapy
Advantage	Familiarity to nursing staff Widespread availability Low cost of disposables Delivery with no anticoagulation feasible	Low cost of disposables Delivery with no anticoagulation feasible Reasonable hemodynamic tolerability	Greater hemodynamic tolerability Ability to adjust prescription rapidly to patient's evolving needs Possibility of improved kidney survival among survivors
Shortcomings	Challenges with hemodynamic instability Limitations of fluid removal	Limited data on appropriate dosing of antimicrobials Lack of randomized controlled trials comparing efficacy with that of other modalities	High cost of disposables Greater logistic complexity Challenging to administer without anticoagulation

actual ultrafiltration volume needs to be 90 mL to achieve a net loss of 50 mL in the subsequent hour. CRRT offers the unique flexibility of tailoring ultrafiltration volumes on an hourly basis according to the variability of intake/losses and the patient's dynamic hemodynamic picture.

SUSTAINED LOW-EFFICIENCY DIALYSIS

Sustained low-efficiency dialysis (SLED), also known as prolonged intermittent renal replacement therapy (PIRRT), is a hybrid modality that uses conventional hemodialysis devices with an extension of therapy to 8 to 12 hours in the hope of achieving some of the putative hemodynamic benefits of CRRT. SLED has gained popularity owing to the absence of a survival benefit with CRRT when compared with IHD, the high costs associated with the application of CRRT,[62,63] and the concomitant desire to provide safe RRT to hemodynamically unstable patients. Initial reports have shown that SLED is associated with reasonable solute control and an achievement of planned ultrafiltration goals.[64-66] SLED was also associated with acceptable hemodynamic stability as compared with CRRT.[66-68] Further practical advantages of SLED include the delivery of therapy without anticoagulation and the potential for administration overnight to minimize interruptions by clinical procedures and investigations that occur during daytime hours.

IMPACT OF MODALITY ON CLINICAL OUTCOMES

CONTINUOUS RENAL REPLACEMENT THERAPY VERSUS INTERMITTENT HEMODIALYSIS

Despite the theoretic benefits attributed to CRRT, randomized trials have not demonstrated enhanced survival in comparison with IHD. The largest of such trials randomly allocated 360 patients (the vast majority of whom were catecholamine-dependent and mechanically ventilated) to IHD or CRRT and demonstrated a 60-day survival of approximately 32% in both groups.[69] A recent meta-analysis of trials and cohort studies comparing the two modalities has shown no benefit of CRRT as compared with IHD (relative risk [RR], 1.00; 95% CI, 0.92–1.09] for mortality and RR of 0.90 [95% CI, 0.59–1.38] for dialysis dependence).[70]

If the putative advantages of CRRT are based on more favorable hemodynamics and the prevention of iatrogenic kidney ischemia, one would expect that better kidney function would be observed in survivors of the acute phase of illness. A meta-analysis of predominantly observational studies has demonstrated a lower risk of dialysis dependence among recipients of CRRT as compared with those who received IHD.[71] A study from Ontario, Canada, has shown that among patients who survived to 90 days after the initiation of acute RRT, the risk of long-term dialysis dependence over 2 years of follow-up was 25% lower among recipients of CRRT than in matched controls whose initial modality was IHD.[72] The possible nephroprotective benefits seen with CRRT are potentially relevant because the cost utility of applying RRT in AKI is closely tied to patients who achieve dialysis independence and survive for longer than 1 year.[73] Although the in-hospital costs of CRRT are higher than for IHD, this initial cost increment may be neutralized if CRRT reduces the risk of long-term dialysis among surviving patients.[62]

CONTINUOUS RENAL REPLACEMENT THERAPY VERSUS SUSTAINED LOW-EFFICIENCY DIALYSIS

The increasing utilization of SLED has been accompanied by limited evidence. A single-center trial randomly assigned 232 patients with AKI admitted to a surgical ICU to SLED (target, 12 hours/session) or CRRT.[74] The primary outcome of 90-day mortality did not differ between the groups (49.6% vs. 55.6% in SLED and CRRT recipients, respectively; P = .43). However, inferences from these results are limited by the small sample size and limited statistical power of this trial. Furthermore, the mean duration of SLED treatments was longer than expected (15 hours), and the mean duration of CRRT sessions was shorter than expected (16 hours). Two studies have evaluated recipients of SLED compared with historical controls comprised of patients treated with CRRT prior to an institutional policy in which SLED replaced CRRT. In one study, the introduction of SLED was associated with improved outcomes compared with patients who were previously treated with CRRT[75] and, in the other, outcomes were comparable.[76] Thus, the limited available data have suggested that SLED is reasonably tolerated by critically ill patients with hemodynamic compromise and appears to be a safe alternative to CRRT.

MODE OF CLEARANCE

As in chronic RRT, solute removal in patients with AKI may be mediated by diffusive and/or convective mechanisms. The relative contributions of diffusive versus convective clearance to the administered therapy may vary and ultimately define the clearance mode—hemodialysis, hemofiltration, or hemodiafiltration. In hemodialysis, a concentration gradient is maintained by the countercurrent flow of dialysate and blood across a semipermeable membrane. The diffusive clearance that ensues is the essence of hemodialysis. With convective clearance, the ultrafiltration of large volumes of plasma water down a pressure gradient forces the concomitant "drag" of solutes through the membrane pores. In hemofiltration, a balanced electrolyte solution devoid of the unwanted solutes then replaces the ultrafiltrate. Hemodialysis and hemofiltration are equally effective in the removal of low-molecular-weight substances (e.g., creatinine, urea, electrolytes). However, because diffusion is size-related, slower moving larger molecules are less efficiently cleared by dialysis. With hemofiltration, in contrast, the clearance of a solute is related to the size of the molecule in relation to the size of the membrane's pores. Thus, depending on the porosity of the membrane, larger molecules, which include potentially toxic cytokines, are more efficiently removed by hemofiltration than by hemodialysis.

Hemofiltration may be provided in tandem with hemodialysis (hemodiafiltration) or as the exclusive mode of clearance. It can be applied in the context of continuous (as in continuous venovenous hemofiltration [CVVH] or continuous venovenous hemodiafiltration [CVVHDF]) or intermittent forms of RRT.[77] Despite its theoretic benefits, there is no evidence that hemofiltration ameliorates clinical outcomes.[78] The largest trial to date on this topic, a comparison of CVVH and CVVHDF, enrolled only 75 patients and did not find a difference in 60-day mortality.[79]

INTENSITY OF RENAL REPLACEMENT THERAPY

Small clinical trials have suggested that the escalation of RRT intensity or dose, defined as increased effluent volume in CRRT[80] or increased session frequency in IHD,[81] could lead to enhanced survival. These findings stimulated two large-scale trials to address definitively whether dialysis intensification could improve survival in AKI. The Acute renal failure Trials Network (ATN) trial randomly assigned 1124 critically ill patients with AKI to two strategies of RRT intensity; in this study, RRT modality varied with in each group, depending on a patient's evolving hemodynamics (Fig. 65.2).[60] Intensive therapy consisted of CVVHDF at 35 mL/kg/hour or SLED, 6 days/week, when the patient was hemodynamically unstable and IHD, 6 days/week, when the patient was hemodynamically stable; the less intensive strategy was CVVHDF at 20 mL/kg/hour or SLED, 3 days/week, when hemodynamically unstable, and IHD, 3 days/week, during periods of hemodynamic stability. For patients receiving IHD or SLED, the dialysis prescription targeted a Kt/V_{urea} of 1.2 to 1.4. The intensive RRT strategy was not associated with lower mortality at 60 days (53.6% vs. 51.5% in the less intensive arm) or a higher likelihood of kidney recovery. The Randomized Evaluation of Normal Versus Augmented Level of Replacement Therapy (RENAL) trial enrolled 1508 critically ill patients with AKI in Australia and New Zealand and compared CVVHDF at

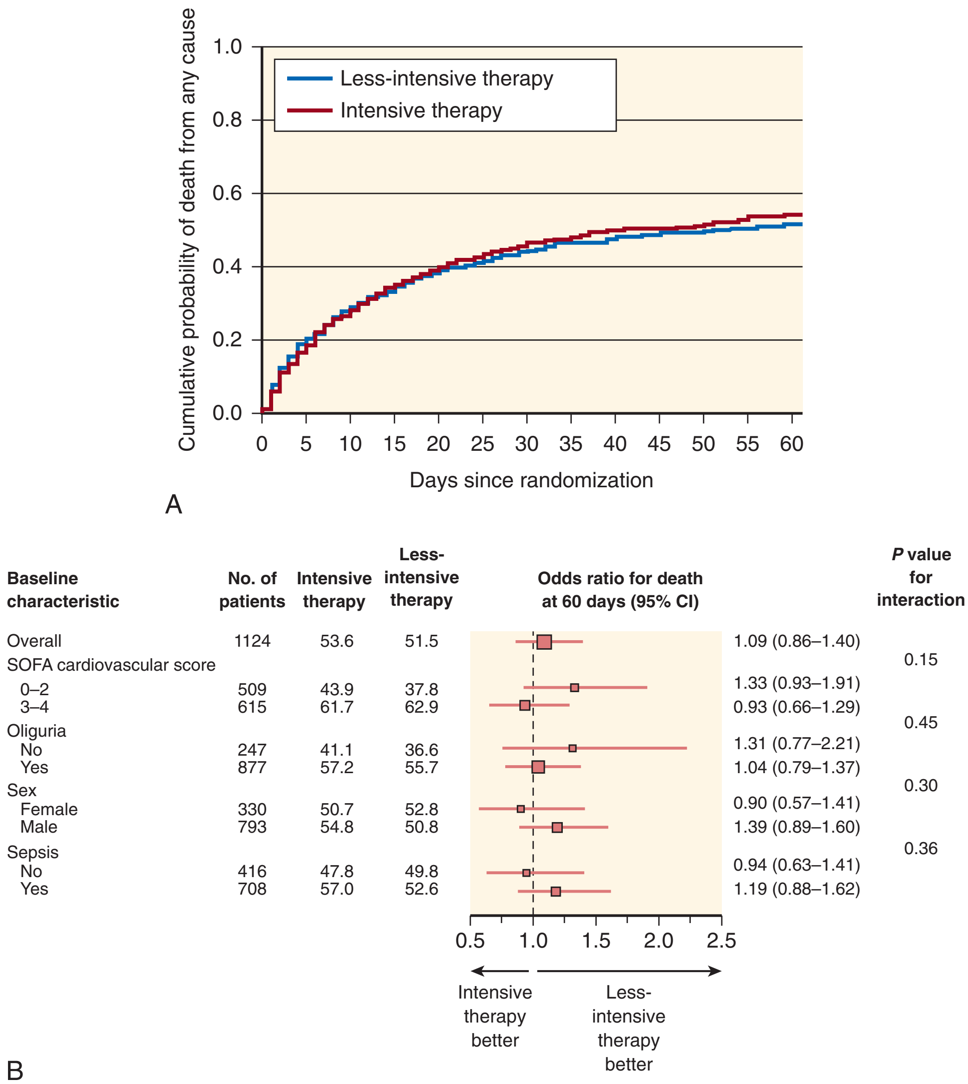

Fig. 65.2 Effect of intensity of renal support in critically ill patients with acute kidney injury. (A) Kaplan-Meier plot of cumulative probabilities of death and odds ratios for death at 60 days, according to baseline characteristics (B) from the VA/NIH Acute Renal Failure Trial Network. (A) shows the cumulative probability of death from any cause in the entire study cohort. (B) shows odds ratios (and 95% confidence intervals [CIs]) for death from any cause by 60 days in the group receiving the intensive treatment strategy, in comparison with the group receiving the less intensive treatment strategy, as well as *P* values for the interaction between the treatment group and baseline characteristics. *P* values were calculated with the use of the Wald statistic. Higher Sequential Organ Failure Assessment (SOFA) scores indicate more severe organ dysfunction. There was no significant interaction between treatment and subgroup variables, as defined according to the prespecified threshold level of significance for interaction (*P* = .10). (From Palevsky PM, Zhang JH, O'Connor TZ, et al. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med.* 2008;359:7–20.)

doses of 40 and 25 mL/kg/hour.[82] Higher CRRT intensity did not confer improved survival at 90 days. No specific advantage of intensive therapy was observed in either trial with respect to prespecified patient subgroups, such as those with sepsis. Studies of ultrahigh CRRT doses (i.e., up to 70–85 mL/kg/hour) have similarly shown no benefit.[25,83] Ultimately, it remains unclear whether the improved control of harmful solutes afforded by high-intensity RRT is counterbalanced by the removal of beneficial solutes such as essential nutrients, endogenous antiinflammatory cytokines, and vital drugs, especially antibiotics. The results of these trials have informed clinical practice guidelines, which currently recommend a target RRT dose of 20 to 25 mL/kg/hour in CRRT and a Kt/V of 3.9/week for patients receiving SLED or IHD.[61]

ANTICOAGULATION

Anticoagulation of the extracorporeal circuit is widely prescribed to counter the propensity of blood to clot when it comes in contact with an artificial membrane in the chronic setting. Other factors that must be considered in the selection of an anticoagulant for RRT include the presence of concomitant indications for anticoagulation, the patient's propensity for bleeding and thrombosis, and the chosen RRT modality.

Regional citrate anticoagulation (RCA) has emerged as an effective and safe anticoagulation strategy and the default strategy for anticoagulation of the CRRT circuit at many centers.[84,85] When infused into the extracorporeal circuit, citrate chelates calcium, a required cofactor in the clotting cascade, and the resulting extracorporeal hypocalcemia (0.25–0.45 mmol/L) prevents circuit clotting. The systemic serum calcium concentration is maintained in the normal range through a concurrent calcium infusion. There are also custom solutions that deliver citrate as part of the replacement solution, thereby obviating the need for a dedicated citrate solution. When compared with unfractionated heparin, RCA confers a lower risk of bleeding.[86,87] There is also accumulating evidence to suggest that RCA prolongs filter life as compared with heparin-based anticoagulation strategies.[88-91] Because citrate is converted to bicarbonate in the liver, RCA often leads to metabolic alkalosis, which can be corrected by reducing the bicarbonate content of the replacement solution or dialysate. In the setting of impaired hepatic function, citrate accumulation may occur, manifesting as a wide anion gap and hypocalcemia. With judicious monitoring, however, RCA can be safely used in some patients with hepatic failure.[92,93]

For patients who require anticoagulation for alternative indications (e.g., mechanical heart valves, deep vein thrombosis), or for those who experience metabolic complications with RCA, unfractionated heparin provides adequate anticoagulation of the RRT circuit. In patients with heparin-induced thrombocytopenia, systemically infused argatroban is an appropriate alternative.[94]

RRT can also be readily administered without anticoagulation. SLED and IHD treatments are more likely to be successfully completed with no anticoagulation owing to the typically higher blood flows and shorter session durations. CRRT may also be administered with no anticoagulation, with an expectation that clotting will necessitate more frequent changing of the extracorporeal circuit. Acceleration of blood flow (e.g., ≥250 mL/min) and the use of hemofiltration with administration of replacement solution in the prefilter position can lower the risk of clotting.

FLUID BALANCE, ULTRAFILTRATION, AND MAINTENANCE OF HEMODYNAMIC STABILITY

The achievement of euvolemia is one of the central goals of any RRT strategy, especially in light of emerging data suggesting the harm of fluid overload.[33] Extracellular volume expansion is often substantial at RRT initiation and has been associated with adverse outcomes.[34-37] In addition, persistent fluid overload while patients are receiving RRT is associated with higher mortality. Although the relationship between fluid overload and death is likely confounded by a variety of factors, an effective ultrafiltration strategy may have many benefits, including reduction in pulmonary edema (which may facilitate ventilator weaning) and mitigation of peripheral edema (which might help the mobilization process).

Effective ultrafiltration entails two challenges. The first is performing a reliable assessment of the patient's volume status. Standard physical examination maneuvers (e.g., assessment of the jugular venous pressure, chest auscultation, peripheral edema) may be difficult to apply in the critical care setting and may be complemented by a search for pulmonary congestion on chest radiographs or information gleaned from a transduced central venous pressure. However, given the limitations of all these tools,[95,96] emerging technologies such as bedside point of care ultrasonography (for assessment of lung fluid[97] and inferior vena cava diameter[98,99]) and bioelectric impedance analysis[100] may prove to be useful adjuncts to standard volume assessment techniques.

A second challenge is the achievement of net fluid removal while avoiding RRT-associated hypotension. Such hypotension occurs, even in patients with bona fide extracellular volume expansion, through inadequate refilling of the intravascular space from the interstitium to compensate for fluid that was removed. The proclivity for hypotension is further enhanced by impairment of vascular tone, which may result from the underlying illness, medications, or dialysis-induced rise in core body temperature.[101] Continuous therapies have been advocated in patients with hemodynamic instability who might derive particular benefit from the low yet consistent ultrafiltration rate afforded by this modality. For patients receiving intermittent modalities, modifications to the dialysis prescription, as described previously, can mitigate iatrogenic hypotension.[102] In addition, novel biofeedback technology, including blood volume monitoring, appears promising but has not been definitively proven to reduce intradialytic hypotension.[103]

VASCULAR ACCESS CONSIDERATIONS

The initiation of acute RRT mandates placement of a dedicated dual-lumen central venous catheter with the goal of sustaining an adequate blood flow with minimal recirculation to ensure adequate solute clearance. To maximize RRT efficiency, the distal catheter tip should reside in a large vein, necessitating the use of a catheter of adequate length. An internal jugular catheter tip should be at the atriocaval junction, thereby requiring a catheter length of 15 to 20 cm; a catheter placed in the femoral veins must reach the inferior vena cava and must have an optimal length of at least 20 to 25 cm.[104] To minimize mechanical and infectious complications of central venous catheters, ultrasound-guided insertion[105] is recommended, and adherence to strict aseptic technique[106] is required.

Although subclavian catheters may be associated with a lower rate of infection,[107] they are generally avoided, owing to the risk of pneumothorax with insertion (especially among patients on mechanical ventilation) and the higher rate of central vein stenosis relative to internal jugular catheters, which might preclude the future creation of a permanent vascular access in the ipsilateral arm in those who will require maintenance hemodialysis.[108,109] The tortuosity of the subclavian vein makes it more likely for the catheter to abut the vessel wall, also increasing the likelihood of endothelial damage and subsequent stenosis.

A randomized clinical trial has shown similar rates of catheter colonization among those receiving internal jugular and femoral catheters overall and no difference in catheter associated-bacteremia.[110] A secondary analysis of these data has suggested a trend for less catheter dysfunction among catheters placed in the right internal jugular vein than in the femoral vein; however, the rate of dysfunction was highest for catheters in the left internal jugular vein.[111] Accordingly, the KDIGO Clinical Practice Guidelines for AKI suggest the right internal jugular vein (which flows in a direct path to the superior vena cava) as the preferred access site for patients with AKI, followed by the femoral veins, left internal jugular vein, and, as a last resort, subclavian vein on the dominant side.[61] Because of their ease of insertion at the bedside, temporary noncuffed catheters are preferred in critically ill patients with AKI. However, even well-functioning, nontunneled catheters should be replaced with cuffed tunneled catheters, which are associated with a lower risk of infection, if a prolonged need for RRT is anticipated.

DRUG DOSING CONSIDERATIONS

In critically ill patients with AKI, medications must be reviewed frequently and meticulously. There are three principal issues of concern:

1. Exacerbation or perpetuation of kidney injury through the administration of nephrotoxins
2. Accumulation of nonnephrotoxic drugs that cause extrarenal toxicity in the setting of impaired kidney function
3. Clearance of vital drugs when patients are receiving RRT

In patients with established AKI, irrespective of cause, avoidance of nephrotoxic medications is a fundamental component of any AKI management strategy. Nephrotoxic agents that are frequently used in the ICU include antimicrobials such as aminoglycosides and amphotericin B, phosphate-based enemas, and radiocontrast agents.[112] Later trials have highlighted the potential nephrotoxicity of hydroxyethyl starches used as volume expanders.[113,114] Alternative agents and imaging strategies should be considered.

Because the glomerular filtration rate (GFR) abruptly declines during AKI, drugs that depend on kidney excretion may rapidly accumulate, with dangerous consequences.[115] Specific examples include morphine, which has an active metabolite that accumulates in the setting of a decreased GFR.[116] The oral anticoagulant dabigatran, a thrombin inhibitor, may rapidly accumulate in the setting of AKI, with dangerous implications for bleeding.[117,118] The accumulation of the sulfonylurea agents used in the treatment of type 2 diabetes mellitus may precipitate hypoglycemia, and metformin might contribute to lactic acidosis.[119] Finally, the accumulation of gabapentin and pregabalin, agents commonly used in the treatment of neuropathy, can cause sedation and myoclonus.[120,121]

Dosing adjustments in the setting of AKI are fraught with challenges because much of the available data about adjustments in drug dosing has been derived from patients with CKD. Even in the setting of a comparable GFR, critical illness and AKI have unique implications for pharmacokinetics. Volume overload may markedly inflate the volume of distribution, necessitating higher loading doses. Once a drug is in the bloodstream, its concentration may be altered by fluctuations in hepatic and kidney metabolism. Finally, drug excretion, predominantly of water-soluble drugs, is impaired in the setting of AKI. Thus, although published recommendations can be used as a starting point for drug dosing, careful monitoring, by following drug levels when available or by clinical assessment for evidence of toxicity, is warranted.

When RRT is initiated, dosing strategies must further account for the extracorporeal clearance of medications. Drug characteristics (e.g., volume of distribution, molecular weight, extent of protein binding), RRT prescription (e.g., blood flow, session duration, proportion of convective and diffusive clearances) and dialyzer features (e.g., porosity or flux, surface area) are important determinants of medication clearance. In some cases, RRT may have a more subtle impact on drug clearance by enhancing extrarenal drug metabolism, perhaps through the removal of uremic toxins.[115] For recipients of intermittent dialysis, medications should be administered after dialysis or a postdialysis supplemental dose should be considered. With the predominant use of high-flux filters, which enhance drug clearance, dosing recommendations that were derived for use with low-flux filters typically need to be increased by 25% to 50%.[115]

The advent of SLED poses a further challenge because extracorporeal clearance is augmented by this modality as compared with IHD. This may result in increased drug removal and compromise blood concentrations of vital medications, including antibiotics. To date, drug dosing guidance with SLED is available for only a few agents.[122] As a general rule, drug doses should probably be higher than those administered to patients receiving standard IHD and, where feasible, their use should be accompanied by careful drug level monitoring.[123] A “top-up” dose following the conclusion of a SLED session is advisable for agents that are likely to be dialyzable. In patients receiving CRRT, total effluent flow and the relative breakdown of convective and diffusive therapy will affect drug removal and subsequent dosing changes. The initiation of CRRT at usual doses, assuming that there are no interruptions to the therapy, will begin to approximate endogenous kidney function, and more frequent drug dosing is generally required than with IHD or SLED. For a more detailed discussion of this topic, the reader is referred to Chapter 61.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 54-year-old man with a history of necrotizing pancreatitis is admitted to the ICU after pancreatic debridement. He remained intubated after the procedure due to hypoxia. Vital signs are temperature, 38.0°C; BP, 102/58 mm Hg on norepinephrine at 10 μg/min; HR, 92 beats per min. He is on Assist Control (AC) ventilation, with a respiratory rate of 12 and tidal volume of 650 mL, positive end expiratory-pressure (PEEP) of 5, FiO_2 1.0. Arterial blood gases (ABGs), 7.36/38/168/25. Chest x-ray shows that the endotracheal tube is in good position and patchy bilateral infiltrates. A lung-protective, low tidal volume ventilation strategy is initiated. What is a common complication of this ventilation strategy that may complicate the management of AKI, if it develops?
   a. Barotrauma leading to pneumothorax
   b. Ventilator dyssynchrony leading to increased intraabdominal pressures and intraabdominal hypertension
   c. Respiratory acidosis due to permissive hypercapnia
   d. Increased intrathoracic pressures leading to decreased renal perfusion

Answer: c.

Rationale: The low tidal volume ventilation strategy targets lower plateau airway pressures and should not exacerbate barotrauma. Ventilator dyssynchrony can be a complication of the lower tidal volumes but, if it occurs, adjustment to the ventilator strategy or increased sedation may be needed, because significant dyssynchrony may obviate the lung-protective benefits. The application of high levels of positive end-expiratory pressure may decrease venous return, but this is not usually clinically significant. In the setting of the acute respiratory distress syndrome and increased dead space fraction, respiratory acidosis due to permissive hypercapnia is relatively common. Metabolic acidosis in the setting of AKI or the underlying critical illness can result in a severe mixed acidosis, and bicarbonate therapy may be indicated.

This applies to questions 2 and 3:

A 65-year-old woman (weight, 60 kg) is admitted to the intensive care unit with urosepsis and associated *Escherichia coli* bacteremia. Her course is complicated by respiratory failure due to acute lung injury and oligoanuric AKI (baseline serum creatinine level, 1.0 mg/dL, currently 5.0 mg/dL; urine output in the past 12 hours, 50 mL). In addition to antibiotics, she is receiving norepinephrine to achieve a mean arterial pressure of greater than 65 mm Hg and is mechanically ventilated, requiring a fraction of inspired oxygen of 40%. Her pH is 7.25, bicarbonate level of 18 mmol/L, and serum potassium level of 4.9 mmol/L. Her cumulative fluid balance since ICU admission is +16 L. The clinical team has elected to commence renal replacement therapy.

2. Which of the following would not be compatible with an optimal strategy of renal replacement therapy?
   a. Intermittent hemodialysis (IHD): initial session 4 hours, with planned ultrafiltration of 4 L
   b. IHD: initial session of 4 hours, with planned ultrafiltration of 1 L
   c. Sustained low-efficiency dialysis (SLED): initial session, 8 hours; blood pump speed, 200 mL/min; planned ultrafiltration of 2 L
   d. Continuous renal replacement therapy (CRRT): net fluid removal of 2 L over subsequent 24 hours

Answer: a.

Rationale: Although IHD may be used in the management of patients with AKI, the ultrafiltration of 4 L (even if the patient is deemed to be markedly volume-expanded) is unlikely to be feasible during a 4-hour session. Intravascular refilling from the interstitium is likely to be compromised, and a strategy of aggressive ultrafiltration would exacerbate hemodynamic instability in a patient who already has septic shock and is vasopressor-dependent. Choices b, c, and d are all acceptable approaches for RRT in this patient. Although SLED and CRRT (choices c and d) would likely be preferable because they would provide a more moderate rate of ultrafiltration and a possibly lower risk of hemodynamic perturbation, there is no evidence to suggest that either of these modalities confers a mortality benefit as compared with IHD. If a decision is made to commence IHD, the ultrafiltration rate should be carefully considered and ideally kept low during the first hemodialysis session, as in choice d. If reasonably tolerated, the ultrafiltration target could be increased for subsequent sessions.

3. A decision is made to commence CRRT. Which of these prescriptions would be considered inappropriate?
   a. Continuous venovenous hemodialysis (CVVHD) with a dialysate flow rate of 1500 mL/hour
   b. Continuous venovenous hemodialysis (CVVHD) with a dialysate flow of 1200 mL/hour
   c. Continuous venovenous hemofiltration (CVVH) with a postfilter replacement flow rate of 1400 mL/hour
   d. Continuous venovenous hemodiafiltration (CVVHDF) with a dialysate flow rate of 500 mL/hour and postfilter replacement flow rate of 500 mL/hour

Answer: d.

Rationale: The optimal recommended intensity or dose of CRRT is 20 to 25 mL/kg/hour, as reflected in the total hourly effluent volume. The hourly effluent volume is the sum of the volume of spent dialysate, the volume ultrafiltered (and replaced 1:1 with convective therapy), and any net ultrafiltration. Because net ultrafiltration may vary from hour to hour, it is preferable not to consider this component when completing the initial CRRT order. This patient, who weighs 60 kg, should have a target dose of 1200 to 1500 mL/hour (i.e., 60 kg × 20–25 mL/kg/hour). As such, choices a, b, and c would each reflect reasonable dosing strategies for CRRT because they would each achieve a total effluent volume in the desired range of 1200 to 1500 mL/hour (20–25 mL/kg/hour). Whereas choices a and b deliver RRT exclusively through hemodialysis (diffusive clearance), and choice c uses hemofiltration exclusively (convective clearance), there is no evidence to suggest superiority for either of these clearance modes. As such, hemodialysis and hemofiltration (or some combination thereof, as is the case with hemodiafiltration) are all acceptable modes of clearance for CRRT. Choice d is inappropriate because the total hourly effluent volume is only 1000 mL/hour (~17 mL/kg/hour) and thus below the recommended dosing target for CRRT.